

Road Accident Risk Predictor

Explainable Risk Scoring for Infrastructure Safety Prioritization

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Summary

Machine learning identifies which road conditions carry disproportionate accident risk, so limited infrastructure safety spending can be targeted rather than spread evenly — with SHAP explanations for every prediction.

Problem

Road safety agencies cannot treat every road segment equally with finite inspection and engineering resources. Road trauma is concentrated in particular conditions rather than spread evenly across a network. The challenge is identifying which conditions carry disproportionate risk so intervention — signage, lighting, speed-limit review — can be prioritized where it has the most potential benefit.

Solution

A Random Forest Regressor trained on 517,754 road-condition scenarios (15 features, including three engineered interaction terms) predicts a risk score for a given road/environment combination and explains *why* via SHAP feature attribution — not just a number, but which factors are driving it (speed, curvature, lighting, weather). Live and publicly deployed at the link above.

Evidence of impact

Ranking all training rows by predicted risk: the **top 10% of road conditions carry 1.93× their proportional share of total predicted risk** (19.3% of total risk mass concentrated in 10% of segments). This is measured directly from the model’s own output, not assumed.

Risk tier	% of segments	Share of total risk	
		mass	Concentration ratio
Top 1%	1%	2.5%	2.50×
Top 5%	5%	11.1%	2.22×

Risk tier	% of segments	Share of total risk mass	Concentration ratio
Top 10%	10%	19.3%	1.93×
Top 25%	25%	41.0%	1.64×

Under scenario-based intervention-effectiveness assumptions (15–30%, sourced from FHWA’s Crash Modification Factors Clearinghouse — a repository of 3,000+ peer-reviewed countermeasure studies), this represents a targeting mechanism with measurable potential benefit. **This is stated as a scenario-based potential impact, not a causal claim** — the model has not been shown to have prevented any crash. Full methodology and worked calculation: `IMPACT.md` in the supporting files.

SDG alignment

Primary — SDG 3.6: halve global deaths and injuries from road traffic accidents.

Secondary — SDG 11.2: safe, affordable, sustainable transport systems.

Mechanism: risk scoring → targeted infrastructure spend → fewer serious injuries. The model does not predict who will crash — it identifies which road conditions carry disproportionate predicted risk.

What is innovative

1. **Predictive** — moves from historical reporting toward forward-looking risk prioritization.
2. **Explainable** — SHAP translates every prediction into interpretable contributing factors, not a black box.
3. **Action-oriented** — connects risk scores to infrastructure intervention prioritization, not just a number on a screen.

Technical evidence

Random Forest Regressor, 200 trees, 517,754 training rows, validation R^2 0.88 — independently reproduced (0.8835), not just claimed. 5-fold CV std ± 0.0003 . No data leakage: no duplicate IDs, and the dataset has no geographic or temporal columns for a random split to leak across. Subgroup performance checked by road type and weather (stable, R^2 0.877–0.885) and by lighting (small honest degradation toward night, R^2 0.857→0.846 — disclosed, not hidden). Full detail: `MODEL_VALIDATION.md`.

Responsible AI

The model uses road and environmental conditions only — no driver identity, ethnicity, or demographic data. Intended for infrastructure planning and safety prioritization, explicitly **not** for individual driver liability, profiling, or enforcement. This is a live decision- support prototype, not a production system with an operator or SLA. Known limitations, including a disclosed subgroup-performance gap in low-light conditions and the boundary of what this

synthetic (Kaggle Playground Series S5E10) dataset can support, are documented in full in `RESPONSIBLE_AI.md`.

Live demo

<https://roadaccident-roshanar-aryal.streamlit.app/>

Hosted on Streamlit Community Cloud’s free tier, which sleeps after inactivity — cold start measured at ~95 seconds. If the link shows “Zzzz... this app has gone to sleep,” click “Yes, get this app back up!” and wait roughly a minute. Screenshots and a walkthrough recording are included in the supporting files as a fallback.

Contents of the supporting zip

- This document (`PROJECT_SUBMISSION.pdf`)
- `screenshots/` — live gauge prediction with SHAP contributions, model info and feature importance
- `walkthrough.mp4` — 60–90s recorded demo (if included)
- `IMPACT.md`, `RESPONSIBLE_AI.md`, `MODEL_VALIDATION.md` — full supporting detail behind the summaries above